Replicating and Extending SimGRL: Enhancing Robustness in Visual RL under Dynamic Scene Perturbations

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Abstract

Visual reinforcement learning (RL) agents are brittle when deployed in environments with dynamic visual perturbations, often failing to generalize against distractors such as moving backgrounds or flickering lights. To address this, the SimGRL framework proposes feature-level frame stacking and Shifted Random Overlay (SRO) to structurally decouple agent dynamics from spurious background correlations (Song et al. 2024). In this work, we replicate these findings to validate their robustness under strict computational constraints. Our results confirm the original authors' claims: even with training limited to 60K steps, SimGRL consistently outperformed the SVEA baseline, achieving a 107% relative improvement in the challenging "Video Hard" setting.

Building on this foundation, we introduced four extensions: Irregular-SRO (I-SRO), Multi-Source SRO (M-SRO), Hybrid-SRO (H-SRO), and Mask-Guided Regularization. These extensions aimed to target residual limitations in the baseline. While I-SRO, M-SRO, and Mask-Guided proved difficult to stabilize within limited training steps, our H-SRO method, which combines spatial perturbations with photometric shifts, emerged as a critical innovation for stability. H-SRO significantly reduced performance variance compared to SimGRL and outperforms the baseline in some cases. This suggests that integrating lighting and exposure invariance is essential for reliable deployment. Our findings highlight a trade-off between augmentation complexity and sample efficiency, suggesting that future robust visual RL should incorporate adaptive curricula to stabilize high-variance perturbations

Introduction

Agents that perceive and act from visual input are central to embodied intelligence. In tasks such as robotic manipulation and autonomous navigation, visual reinforcement learning enables agents to learn behaviors directly from raw pixel observations, eliminating the need for handcrafted state representations. These learned behaviors are represented by control policies, which map visual inputs to actions that seek to maximize long term reward. Despite algorithmic progress, such policies remain highly sensitive to changes in visual input between training and deployment. Variations in background, lighting, or dynamic elements can significantly degrade performance, limiting real world reliability.

SimGRL is a simple and effective framework designed to improve generalization in visual reinforcement learning under dynamic visual perturbations. It identifies two key sources of brittleness. First, imbalanced saliency arises in partially observable environments, where frame stacking causes agents to overfocus on recent, noisy frames while ignoring informative past context. For example, in a Cheetah task where a simulated quadruped learns to run forward, an agent may respond to background flickers rather than stable motion cues. Second, observational overfitting occurs when agents rely on background features correlated with reward but unrelated to task dynamics. SimGRL addresses these problems using feature level frame stacking to balance temporal attention, and Shifted Random Overlay (SRO) to decorrelate spurious background features. These issues are illustrated in Figure 1. Together, these methods improve generalization on the DMControl Generalization Benchmark with minimal added complexity.

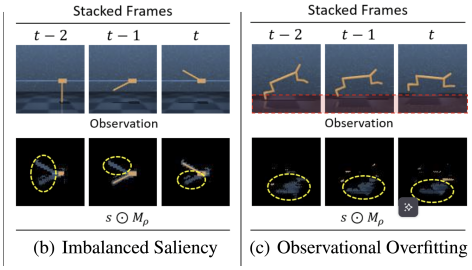


Figure 1: Original figures from SimGRL paper: (b) Imbalanced saliency: attention is biased toward recent frames. (c) Observational overfitting: attention is misdirected to background features. Visualized with attribution maps from a trained DrQ critic (Song et al. 2024)

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In this paper, we replicate and extend SimGRL. Our replication on the Walker Walk task confirms its advantages over the SVEA baseline under Train, Video Easy, and Video Hard evaluations, using only 60K training steps, far fewer than the original 500K. Building on this foundation, we propose four targeted extensions: **I-SRO** breaks background motion coherence; **M-SRO** varies background appearance per frame; **H-SRO** adds photometric noise such as color and blur; **Mask-Guided** penalizes off object saliency using supervised attention.

While all extensions perform worse on average under limited training, we expect I-SRO and M-SRO to improve with longer horizons due to their regularization effects. H-SRO achieves the most consistent performance across seeds, and Mask-Guided shows strong peak performance and potential for object aware control. All developed codes are released at https://github.com/xiyanzho/498research.

Related Work

Generalization in visual RL has been widely studied through data augmentation. Early work like RAD (Laskin, Srinivas, and Abbeel 2020) and DrQ (Yarats, Kostrikov, and Fergus 2021) demonstrated that simple transformations (e.g., random crops, shifts) improve sample efficiency. However, these techniques often fail to prevent agents from relying on spurious background dynamics. To address stability issues under strong augmentations, SVEA (Hansen, Su, and Wang 2021) introduced a dual-view framework using both augmented and unaugmented observations to stabilize critic learning. Similarly, CG2A (Liu et al. 2023) proposed a gradient-agreement mechanism to reconcile conflicting gradients from multiple augmentations.

A complementary approach incorporates saliency to achieve object-centric learning. SGQN (Bertoin et al. 2023) uses self-supervised saliency prediction to focus the value network on relevant regions, while ViSaRL (Liang, Thomason, and Bıyık 2024) guides agents using human attention maps. More recently, SCPL (Sun et al. 2025) enforces saliency-invariant consistency to achieve strong zero-shot transfer. SimGRL aligns with these strategies by combining feature-level frame stacking with dynamic background perturbations via Shifted Random Overlay (SRO). Our extensions build on this foundation by exploring irregular motion, multi-source overlays, and mask-guided regularization.

Background

Problem Formulation & Algorithm Based on (Song et al. 2024)'s work on SimGRL, we operate in a Partially Observable Markov Decision Process (POMDP) setting where agents learn from high-dimensional pixel observations without explicit state information. We employ Soft Actor-Critic (SAC) (Haarnoja et al. 2018), a sample-efficient maximum entropy algorithm, as the backbone for our experiments.

Feature Level Stacking Instead of stacking raw images, the agent processes the current frame and past frames independently through the same shared encoder. This treats each moment in time as a separate snapshot, allowing the encoder to identify and extract relevant features from the noisy background before any temporal mixing occurs

Benchmark & Metrics We evaluate on the DeepMind Control Generalization Benchmark (DMControl-GB) (Hansen, Su, and Wang 2021), which tests agents on Video Easy and Video Hard variants containing diverse background distractions. To measure how much the model identifies the task object's pixels across stacked frames, (Song et al. 2024) introduced Task Identification Score (TID Score), and higher TID score indicates the agent is focusing on the robot rather than the background. The formula for TID score calculation is:



Figure 2: Original figure from SimGRL paper: Examples of augmented images using shifted random overlay (SRO) (Song et al. 2024)

$$TID_S = \sqrt{\frac{(N_{objM})^2}{N_{obj} \times N_M}}$$

where N_{obj} denotes the total number of pixels belonging to the task object in the input image, and M_ρ represents the attribution mask generated using the quantile threshold ρ. Consequently, N_M is the total pixel count of the mask M_ρ, while N_{objM} represents the number of object pixels successfully captured within that mask.

Shifted Random Overlay (SRO) To prevent overfitting to static backgrounds, SimGRL utilizes SRO during training. SRO samples background images from the Places365 dataset (Zhou et al. 2018). For an episode, it defines a smooth trajectory by cropping a region with offsets ($\triangle h$,$\triangle w$) relative to a base coordinate. The background is alpha-blended ($\alpha = 0.5$) to the foreground (DM control background and object movement). A higher value of α increases the influence of the background, whereas a lower value keeps the original DMControl frame more dominant (Figure 2). Crucially, SRO maintains temporal coherence—the background shifts smoothly across frames, mimicking camera panning.

Methodology, Results, and Discussion

To replicate SimGRL, we utilized database, code, and environment setup from (Song et al. 2024) and focus on the Walker Walk task. We determined to compare SimGRL with SVEA, which is the model SimGRL built upon. Comparing SimGRL and SVEA isolates the effect of SRO and feature level stacking frame without introducing architectural differences.

Ideally, our replication would mirror the original study's experimental design, where agents were trained for 500K steps until performance saturation. However, due to the strict computational limits of the Great Lakes cluster, replicating the full trajectory was infeasible. Consequently, we adjusted our scope to focus on the following two specific goals. First, we evaluate whether SimGRL outperforms SVEA at 60K training steps across Train, Video Easy, and Video Hard because that is maximum the number of step we can run. Though with fewer steps, consistent superiority at this early stage would still support the claim SimGRL is better than SVEA. Second, we compare our learning trends in the 0–60K range with the early-stage portions of the training curves proposed by (Song et al. 2024). Similarity in early-phase trends can also provide meaningful evidence that our reduced-budget setup captures the qualitative behavior of

SimGRL.

These objectives are sufficient to establish SimGRL's superiority under visual perturbations. By confirming the relative performance gap and divergent learning dynamics at just 60K steps, we effectively validate the paper's core claims without needing to reproduce the absolute final scores.

Eval Mode	SVEA	SimGRL	Δ
Train	556 ± 79	684 ± 81	128 (23%)
Video Easy	524 ± 62	664 ± 79	140 (27%)
Video Hard	269 ± 71	557 ± 71	288 (107%)

Table 1: Replication results on Walker Walk (DMControl-GB). Means $\pm$ standard deviations over 3 seeds. $\triangle$ shows SimGRL's absolute and relative improvement over SVEA.

Across all evaluation settings, our replication confirms that SimGRL consistently demonstrates stronger generalization than SVEA, with improvements observed not only in the training environment but also in both test distributions. As summarized in Table 1, the performance gap grows progressively larger as the visual perturbations become more severe.

Under the Train and Video Easy conditions, SimGRL offers moderate gains over SVEA, roughly 20–30% on average, which indicates more stable representation learning even before encountering significant distribution shift. However, the advantage becomes most evident in the Video Hard setting, where backgrounds are fully replaced with dynamic videos. In this challenging regime, SimGRL more than doubles SVEA's reward (107 % relative improvement), demonstrating that its architectural and augmentation strategies meaningfully improve robustness to background dynamics.

This cross-seed consistency and the widening gap under harder perturbations highlight a key outcome of our replication: SimGRL's generalization benefit is not merely a small optimization gain, but a structural improvement that scales with environment difficulty. In other words, as the shift becomes more extreme, SimGRL degrades far more gracefully than SVEA—validating the original paper's claim that it mitigates overfitting to spurious background features.

These results confirm that our replication successfully captures the central empirical claims of the original SimGRL paper. Across all three evaluation modes, SimGRL consistently outperforms SVEA at 60K training steps, matching the reported trend that SimGRL provides stronger robustness under increasingly difficult visual perturbations. This pattern appears both in the aggregated scores (Table 1) and in the per-seed learning curves (Figure 3), where SimGRL improves more steadily and achieves substantially higher performance on the Video Hard benchmark. While our absolute performance values are lower than those reported by the authors, their original results show that SVEA reaches 819 ± 71 on Video Easy and 385 ± 63 on Video Hard in the Walker Walk task, whereas SimGRL achieves 910 ± 21 and 773 ± 31 under the same settings. These numbers provide the reference scale for comparison, and the difference does not indicate a failure of replication. The discrepancy is expected given that the original experiments train for

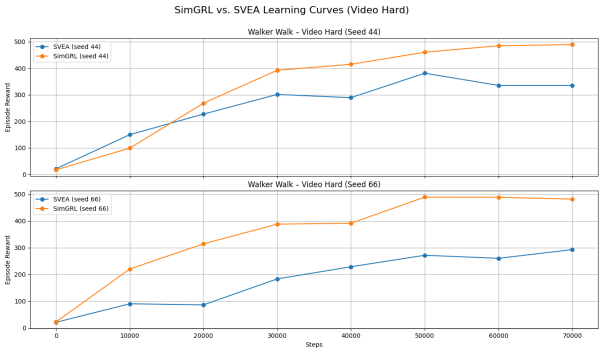


Figure 3: Training curves for SVEA and SimGRL on Walker Walk (Video Hard) for seeds 44 and 66. SimGRL learns faster and achieves consistently higher returns under dynamic background shifts.

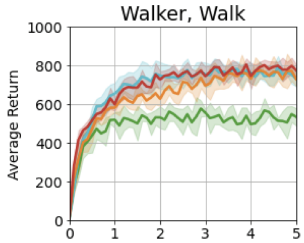


Figure 4: Training curves from the original SimGRL paper on Walker Walk (Video Hard). The x-axis is in units of 10^5steps. SimGRL variants outperform SVEA and show more stable learning (Song et al. 2024).

500K steps with five seeds, whereas our setup is limited to 60K steps and three seeds due to the 8-hour job constraint on Great Lakes. Under such constraints, lower numerical scores are unavoidable. What matters for replication is that the *behavioral trends* match the original paper (Figure 4), and our results clearly reflect this: the relative ordering between SimGRL and SVEA, the widening performance gap as perturbation difficulty increases, and the early-stage learning dynamics shown in our training curves closely mirror the patterns reported in the original work. Also, the TID score, TID variance, and Generalization Gap's patterns align with the paper (Figure 5). Differences in the final reward values caused by computational limits do not impact this alignment of trends. Overall, both evaluation components of our replication succeed. First, SimGRL maintains a clear performance advantage over SVEA across all evaluation modes even at 60K steps, supporting the primary comparative claim of the original work. Second, the learning trends in our training curves closely resemble the early-stage behavior reported in the SimGRL paper, including the widening performance gap as perturbation difficulty increases. Together, these findings indicate that our reduced-budget replication faithfully reproduces the essential qualitative dynamics of SimGRL, even if it does not match the full-scale numerical results from the 500K-step experiments.



Figure 5: Comparison of SVEA vs. SimGRL across (left) TID variance, (middle) TID score, and (right) generalization gap.

Extensions

To evaluate our extensions, we used a unified training and testing setup. Due to computational limits on Great Lakes, we trained all models for 60K steps rather than the full 500K used in SimGRL, with evaluations every 10K steps. After training, the final checkpoint was assessed over 30 episodes across Train, Video Easy, and Video Hard modes. This reduced budget means our models may not reach full SimGRL convergence. Therefore, our results emphasize relative performance under a fixed, limited compute setting rather than absolute replication of SimGRL's original outcomes.

Building on SRO, our extensions target two directions highlighted in prior work: (1) increasing temporal/appearance irregularity (I-SRO, M-SRO, and H-SRO)) and (2) enforcing object-centric representations (Mask-Guided).

Extension 1: I-SRO (Irregular-SRO)

The standard SimGRL framework utilizes Shifted Random Overlay (SRO) to mitigate observational overfitting by injecting background motion. However, SRO relies on a temporally coherent shift vector; the background moves smoothly in a single direction (dh, dw) across the episode. We hypothesize that this smoothness introduces a spurious correlation: the agent may learn to rely on the consistent optical flow of the background to estimate its own ego-motion, rather than focusing solely on the target agent. (Song et al. 2020) In real-world dynamic scenes, visual noise is rarely so predictable (Tobin et al. 2017).

Although the authors of SimGRL mention an "Irregular-SRO" (I-SRO) variant in their appendix, they did not evaluate it within the feature-level frame stack context. We propose that by breaking this temporal coherence, we can force the agent to disregard background features entirely, as they become mathematically unpredictable noise.

I-SRO modifies SimGRL's augmentation pipeline by sampling a new random shift vector per frame, in which l is the maximum shift possible:

$$(\triangle x_i, \triangle y_i) \sim \text{Uniform}\left([-l, l]\right)$$

The cropped coordinates (h_i, w_i) for frame i are then calculated relative to the base coordinate (h_0, w_0).

$$(h_i, w_i) = (h_0 + \triangle x_i, w_0 + \triangle y_i)$$

Thus each frame receives a different perturbation and produces temporal discontinuity in overall backgrounds while preserving the spatial statistics of the overlay images.

As shown in Table 2 and 3, the I-SRO method underperformed relative to the baseline SimGRL (-12.8% for Video Easy and -13.0% for Video Hard in average). While initially counter-intuitive, this result highlights the trade-off between robustness and sample efficiency. We interpret this performance drop through two primary factors:

Loss of Relative Motion Estimation: The baseline SRO's coherent background motion likely acted as a stabilizing visual cue, allowing the agent to infer its own movement relative to the background. By removing this coherence, I-SRO increased the difficulty of the state estimation task (Song et al. 2020).

Variance and Sample Efficiency: I-SRO introduces significantly higher variance into the observation space than standard SRO. In Reinforcement Learning, higher variance in the input stream necessitates a larger number of training samples to converge to an optimal policy. Under our limited budget of 60K steps, the agent lacks the capacity to adapt to the increased visual randomness. Thus, while I-SRO may prevent overfitting, its benefits are likely to emerge only under longer training horizons.

This suggests that while I-SRO eliminates the agent's reliance on background consistency, the increased task difficulty requires a longer training horizon to yield performance benefits.

Extension 2: M-SRO (Multi-source SRO)

While I-SRO disrupts the temporal coherence of the background's motion, it still utilizes a single static image for the duration of an episode. Consequently, the background retains textural consistency, including the visual statistics, style, and patterns of the background. This may allow the agent to overfit to specific texture statistics or low level visual features of the static background image (Geirhos et al. 2019).

To rigorously test the agent's ability to ignore visual distractors, we introduce Multi-Source SRO (M-SRO). This method aims to break temporal coherence in both motion and appearance. A similar concept, termed "Arbitrarily Stacked Random Overlay (A-SRO)", was briefly introduced in the SimGRL appendix (Song et al. 2024) but lacks thorough evaluation within the feature-level frame stack context.

M-SRO modifies the augmentation pipeline to maximize appearance variance. Instead of sampling one background image per episode, M-SRO samples a distinct background image from the natural image pool for every individual frame. For each time step, a new image is sampled from the Place365 dataset. The image is resized to match the observation resolution. Then, the background is blended with the foreground video with a blending factor $\alpha = 0.5$.

The quantitative results for M-SRO (Table 2) is very similar to I-SRO, with scores -12.4% and -11.0% for Video Easy and Video Hard compared to SimGRL. This is not unexpected when imposing such distractions to vision RL models (Stone et al. 2021), and can be explained in the same way

as I-SRO. The M-SRO environment is arguably the most difficult of the proposed variants, and the agent was likely unable to filter out the intense background noise within the constrained 60K-step budget.

This is further proved as we observed that reducing the blending factor from $\alpha = 0.5$ to $\alpha = 0.3$, which made background less dominant, resulted in higher rewards. (Table 6 in Appendix) This indicates that the agent is highly sensitive to the dominance of the background signal in the observation and strongly supports our hypothesis that I-SRO and M-SRO underperformed because they overwhelmed the agent's encoder with intensified background variance.

Extension 3: H-SRO (Hybrid SRO)

While SimGRL effectively utilizes SRO to distract the agent, real-world visual variation is not limited to movement. Environments often exhibit photometric shifts, such as changes in lighting, color drift, camera exposure adjustments, and motion blur.

Prior work in visual reinforcement learning, such as RAD, has demonstrated that augmenting observations with color jitter, grayscale conversion, and blur significantly improves robustness by forcing the agent to learn features invariant to appearance statistics (Laskin, Srinivas, and Abbeel 2020). We hypothesized that combining these photometric perturbations with SRO's spatial perturbations would yield a "Hybrid" approach (H-SRO). By simultaneously randomizing motion and appearance, the agent is encouraged to learn a representation that is invariant to both dynamic background shifts and low-level pixel statistics.

To implement H-SRO, we inject a pipeline of lightweight appearance transformations prior to the standard SRO overlay process. For each incoming frame in the stack, we independently apply one or two augmentations are sampled from a set of standard transforms, including color jitter (randomizing brightness, contrast, saturation, hue), grayscale conversion, and Gaussian blur. Then, distinct from the large-scale SRO background shifts, we introduce an intensified pixel-level shift with maximum shift 8 pixels, which is handled by periodic wrapping, which was proved to be useful in enhancing model robustness (Laskin, Srinivas, and Abbeel 2020).

These operations are applied independently per frame. This ensures that appearance variations fluctuate across time, disrupting fine-grained pixel statistics while preserving the overall structure of the scene.

Although H-SRO's average performance is slightly lower than SimGRL (Table 3), it exhibits markedly higher stability across random seeds. While SimGRL achieves strong mean scores, its standard deviations range from 71 to 81 across evaluation settings, indicating considerable sensitivity to initialization. In contrast, H-SRO maintains much lower variance, with standard deviations of 24 in Train, 25 in Video Easy, and 22 in Video Hard. For example, under Seed 1 where SimGRL and most other variants show degraded performance, H-SRO remains relatively consistent. This suggests that H-SRO is more robust to randomness in policy initialization and early training dynamics. In scenarios where computational budgets limit retraining or parameter tuning, such as in real time systems, field robotics, or socially impactful deployments with strict resource or safety requirements, performance consistency may be more important than achieving the highest possible return. Under these conditions, we consider H-SRO a strong alternative to SimGRL due to its reliable and stable behavior.

Extension 4: Mask-Guided Regularization

Although (Song et al. 2024) uses TID only as an evaluation metric applied after training in SimGRL, we extend its role by incorporating it directly into the training process. The intuition is that the ground-truth object mask indicates the region that should guide the model's attention. Therefore, we think it is a good idea to add TID score into the loss function and use TID score to guide the model to focus on the correct objects. Our Mask-Guided extension uses this information by adding a regularization term that penalizes saliency outside the object region, encouraging the critic to focus on the agent's body rather than background variations.

To keep the mask-guided training effective while limiting additional cost, we introduced three hyperparameters.

- `mask_reg_lambda`: sets the strength of the mask-based penalty.
- `mask_reg_freq`: controls how often saliency maps are computed.
- `mask_reg_quantile`: Each pixel implies a gradient that represents its sensitivity regarding changing the Q-value. The quantile determines the top percentage of pixels, ranked by gradient magnitude, that are included in the saliency mask.

We compared the Mask-Guided extension to the SimGRL baseline on DMControl-GB (Hansen, Su, and Wang 2021). As shown in Table 3, the Mask-Guided method achieved a lower average performance compared to SimGRL, such as 482 ± 171 versus 557 ± 71 in the "Video Hard" setting. The significantly higher standard deviation indicates that the added regularization made the optimization process less stable, causing some training runs to fail. However, looking closely at individual seeds in Table 2 reveals the method's potential. In the "Video Easy" task, Seed 0 of the Mask-Guided model achieved a score of 777, surpassing the SimGRL baseline of 740. Similarly, in "Video Hard," Seed 0 scored 681, again outperforming the baseline. These peak scores suggest that while the current regularization parameters make training harder to tune, the Mask-Guided approach can produce superior policies when the training stabilizes, potentially outperforming augmentation-only methods with further hyperparameter optimization.

Beyond these empirical observations, the Mask Guided approach is conceptually well grounded. It encourages the critic to rely on object centric information, introducing an inductive bias that matches the structure of DMControl GB tasks, where the agent's body contains most of the task relevant signals. This bias becomes especially important in environments with strong visual distractions, where augmentation alone may still allow the model to pick up background specific patterns. With additional training steps or better tuning of the regularization strength and frequency, the stability

Model	Seed	SimGRL	ISRO	HSRO	MSRO	MG
Train	0	760	653	664	657	772
	1	572	509	633	559	508
	2	719	632	604	567	436
Video Easy	0	740	631	643	652	777
	1	556	505	639	558	507
	2	697	601	586	536	427
Video Hard	0	630	529	558	569	681
	1	460	410	563	470	426
	2	580	514	510	447	338

Table 2: Performance across seeds for Train, Video Easy, and Video Hard. MG stands for Mask-Guided.

issues observed in our current runs may be reduced, helping the model learn cleaner and more reliable object focused features. Overall, this form of attention guidance complements augmentation by giving the model a clearer indication of which regions are relevant during training. It also has potential societal implications, as object aware mechanisms may support more interpretable and responsible deployment in domains where transparency and safety are essential, such as healthcare or assistive robotics.

Eval Mode	Model	Mean $\pm$ Std	Δ (%)
Train	SimGRL	684$\pm$81	0 (0%)
	I-SRO	598$\pm$64	-86 (-12.5%)
	H-SRO	634$\pm$24	-50 (-7.3%)
	M-SRO	594$\pm$44	-89 (-13.1%)
	MG	572$\pm$168	-112 (-16.3%)
Video Easy	SimGRL	664$\pm$79	0 (0%)
	I-SRO	579$\pm$54	-85 (-12.8%)
	H-SRO	623$\pm$25	-42 (-6.3%)
	M-SRO	582$\pm$50	-82 (-12.4%)
	MG	570$\pm$167	-94 (-14.2%)
Video Hard	SimGRL	557$\pm$71	0 (0%)
	I-SRO	484$\pm$53	-72 (-13.0%)
	H-SRO	544$\pm$22	-13 (-2.3%)
	M-SRO	495$\pm$53	-61 (-11.0%)
	MG	482$\pm$171	-75 (-13.5%)

Table 3: Mean $\pm$ standard deviation across 3 seeds comparing all SRO and Mask-Guided to SimGRL (baseline). MG stands for Mask-Guided.

Conclusions

In this work, we replicated the SimGRL framework based on the paper *A Simple Framework for Generalization in Visual RL under Dynamic Scene Perturbations* (Song et al. 2024), validating its core claim that feature level stacking and SRO enhance visual robustness. Despite a constrained training budget of 60K steps, SimGRL consistently outperformed the SVEA baseline. This advantage was most pronounced in the Video Hard setting, where SimGRL achieved a 107 percent relative improvement, supporting the view that its inductive biases enable faster adaptation to background distractors.

Among our extensions, H-SRO demonstrated the highest stability across seeds. While it did not achieve the highest average reward, it consistently produced strong performance under varying initializations, suggesting robustness to randomness in policy optimization. This makes it well suited for real world applications where retraining is costly or infeasible. In contrast, I-SRO and M-SRO underperformed under limited training, likely due to their higher input variability. However, their designs remain promising for chaotic environments with irregular motion or diverse distractors. These results indicate that such methods may require longer training horizons to realize their regularization benefits. The mask-guided extension, although unstable on average, reached the highest peak performance in several runs, indicating its capacity to produce strong policies under favorable conditions. Separately, its object-focused regularization may offer value in domains that require interpretability or spatial grounding, such as medical diagnostics or human-assistive systems.

Future research could adopt adaptive curricula that gradually introduce strong augmentations such as I-SRO and M-SRO. Starting from clean inputs and increasing perturbation over time may help the agent develop stable control before confronting highly variable observations. This could mitigate early collapse under chaotic inputs. Larger scale experiments with extended training budgets (up to 500K steps) and more seeds (10 or more) are necessary to fully assess the potential of these extensions. In parallel, the mask-guided approach warrants further study in domains where it is important to understand and visualize which features influence the agent's behavior. Finally, testing across more diverse control environments, such as high speed locomotion in Cheetah Run, would improve the generality of our findings beyond Walker Walk.

Societal Impact

Advancing robust visual reinforcement learning is increasingly important for deploying embodied AI systems beyond controlled laboratory settings. By improving an agent's ability to remain stable under visual perturbations, our work has potential implications for autonomous vehicles, assistive devices, and service robots. Our results on H-SRO emphasize the stable representations under limited compute budgets, a critical requirement for edge-deployed systems where retraining or extensive fine-tuning is impractical. Likewise, our exploration of Mask-Guided regularization introduces an object-centric inductive bias that may support more interpretable decision-making, which is desirable in sensitive domains such as healthcare or human–robot interaction.

At the same time, our techniques that enhance robustness and invariance can be misused. Stronger perception models may increase the persistence and reliability of automated surveillance systems or reinforce biased behavior when deployed in opaque policy-enforcement tools. These risks highlight the need for ethical evaluation, transparency, and development of guidelines to ensure models are used responsibly. Educational tools may also benefit from models that clearly show what they are attending to, improving transparency and trust.

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Individual Contributions

Yiyang Fan I served as the project manager, setting the overall timeline, dividing tasks, and coordinating weekly goals for the team. I created the initial paper outline, specifying the content and structure of each section, and later reviewed all writing drafts, providing edits and revision guidance to ensure clarity and cohesion. In addition to management and writing responsibilities, I implemented and tested M-SRO extensions, including their training and evaluation pipelines on the Great Lakes cluster. I also contributed to interpreting results and shaping the discussion and conclusion sections.

Xiyan Zhou I implemented the H-SRO extension, including the full augmentation pipeline and its integration into the SimGRL framework. I conducted training and evaluation experiments to identify effective hyperparameters and generate the results reported in the paper. I also contributed substantial writing to the background and data-augmentation extension sections and assisted in revising and refining the overall paper.

Yihan Lei I implemented the Mask-Guided Regularization module, including saliency-based loss computation and the design of associated hyperparameter controls. I ran the full SimGRL replication experiments under the project's compute constraints, producing the baseline results used to evaluate our extensions. I also assisted with debugging the training scripts and verifying the correctness of reproduced models. In addition, I contributed to writing the Replication section and the Mask-Guided extension section of the paper.

Mike Xiangyu Cai I implemented the I-SRO extension, including its augmentation logic and integration into the SimGRL training pipeline. I also contributed substantial writing to the Abstract, Introduction and Conclusion sections of the paper, helping frame the motivation and synthesize the project's findings. In addition, I assisted with final manuscript revisions to ensure clarity, organization, and coherence across sections.

Yiming Feng Did a well rounded research in RL: SAC, Q-learning and Computer Vision: frame-stack, feature extraction, to write the introduction and background section of this paper. Managed and updated timeline each week, and contributed in the code environment setup.

Xingge Li I contributed by reviewing the final draft of the paper, checking for consistency across sections, tables, and figures. I provided feedback on clarity, organization, and presentation to help refine the overall quality of the manuscript. Assisted with environment configuration, running training jobs, and benchmarking results across Train, Video Easy, and Video Hard settings.

Model	Seed	SVEA	SimGRL	ISRO+MG	MSRO+MG
Train	0	667	760	581	628
	1	493	572	517	494
	2	507	719	544	473
Video Easy	0	611	740	483	632
	1	475	556	503	475
	2	486	697	503	460
Video Hard	0	368	630	67	575
	1	202	460	136	341
	2	237	580	106	393

Table 4: Performance of SVEA, SimGRL, ISRO with Mask-Guided, and MSRO with Mask-Guided on Walker Walk across three seeds for Train, Video Easy, and Video Hard. MG stands for mask-guided.

Appendix

Environment Configuration

To replicate the core findings of SimGRL, we first downloaded the official implementation released by the authors and reconstructed their software environment to match the original setup as closely as possible. We installed the same versions of PyTorch, DMControl, and all augmentation-related dependencies, ensuring that our package configuration aligned with the authors' repository. Because SimGRL uses the Places365 dataset to generate natural-image background perturbations, we downloaded the same dataset and reproduced their preprocessing pipeline, including image resizing, uniform sampling, and random frame selection. We also configured DMControl-Generalization (DMControl-GB) according to the specifications in the paper. DMControl-GB is an extension of the standard DMControl benchmark designed specifically to test visual generalization: instead of evaluating agents only in the environment they were trained on, it introduces new appearance variations such as video backgrounds, natural textures, and visual shifts to measure how well a policy transfers to unseen visual conditions. In our setup, we matched the observation resolution, frame stacking, replay buffer structure, and the actor–critic architecture used in the original implementation.

Integrated Extension: Mask-Guided SRO Extensions

We explored combining the Mask-Guided (MG) method with three SRO variants: I-SRO, H-SRO, and M-SRO, because these extensions operate on different components of the training pipeline. MG modifies the loss function by introducing an attention-guidance penalty, whereas SRO methods perturb the background distribution through spatially shifted natural image overlays. Since one modifies the optimization objective and the other diversifies visual observations, we expected their mechanisms to be complementary and potentially yield stronger generalization.

Table 4 reports the performance of the integrated variants. The ISRO+MG combination performed poorly despite ISRO preserving the agent and perturbing only the

Eval Mode	Model	Mean ± Std	Δ (%)
Train	SimGRL	684±77	0 (0%)
	M-SRO ($\alpha = 0.5$)	594±42	-90 (-13.1%)
	M-SRO ($\alpha = 0.3$)	632±59	-52 (-7.6%)
	ISRO+MG	547±26	-137 (-20.1%)
	MSRO+MG	532±79	-152 (-22.2%)
Video Easy	SimGRL	664±76	0 (0%)
	M-SRO ($\alpha = 0.5$)	582±47	-82 (-12.4%)
	M-SRO ($\alpha = 0.3$)	622±47	-42 (-6.3%)
	ISRO+MG	496±12	-168 (-25.3%)
	MSRO+MG	522±91	-142 (-21.3%)
Video Hard	SimGRL	557±69	0 (0%)
	M-SRO ($\alpha = 0.5$)	495±62	-62 (-11.1%)
	M-SRO ($\alpha = 0.3$)	526±34	-31 (-5.6%)
	ISRO+MG	103±29	-454 (-81.5%)
	MSRO+MG	436±118	-121 (-21.7%)

Table 5: Mean performance and percentage change from SimGRL across seeds.

background. A likely explanation is that ISRO introduces highly irregular background shifts across consecutive frames, which amplifies the variance of the MG saliency gradients. Because MG relies on stable gradient patterns to form reliable mask-alignment penalties, these fluctuations can interfere with optimization and lead to unstable learning.

The HSRO+MG variant also struggled. We only trained 20K steps in 8 hours. HSRO applies structured, regularly shifted overlays that are computationally heavier than ISRO, increasing the cost of each update. When combined with MG's saliency computation, the overall training process becomes significantly slower, limiting the number of effective optimization steps available under a fixed training budget.

Among the three combinations, MSRO+MG achieved the most consistent performance. MSRO perturbs backgrounds in a more uniform manner, reducing frame-to-frame variability and allowing the MG penalty to be applied more reliably. However, the added regularization still complicates optimization, preventing the integrated method from surpassing the standalone MSRO or SimGRL baselines.

Nevertheless, integrating MG with SRO methods remains a promising direction. MG offers an explicit signal that encourages object-centered reasoning, while SRO improves robustness to visual distractors. Aligning mask-guided regularization with more stable augmentation schedules, or allocating larger training budgets, may allow these complementary forms of supervision to reinforce each other more effectively, enhancing generalization in visually complex environments.

Changing the Background Mixing Coefficient α

The Shifted Random Overlay (SRO) module in SimGRL blends each DMControl observation with a natural-image background using a mixing coefficient α. In the original implementation, the blended image is computed as

$$x_{\text{blend}} = \alpha \cdot x_{\text{bg}} + (1 - \alpha) \cdot x_{\text{obs}}, \quad \alpha = 0.5.$$

A higher value of α increases the influence of the background, whereas a lower value keeps the original DMControl frame more dominant. The choice of $\alpha = 0.5$ provides strong but stable perturbations, especially because SRO maintains temporal coherence by smoothly shifting background frames to mimic camera panning.

Dataset	Seed	M-SRO ($\alpha = 0.5$)	M-SRO ($\alpha = 0.3$)
Train	0	657	698
	1	559	618
	2	567	580
Video Easy	0	652	686
	1	558	606
	2	536	574
Video Hard	0	569	564
	1	470	518
	2	447	497

Table 6: Comparison of M-SRO performance with $\alpha = 0.5$ and $\alpha = 0.3$ across three seeds.

To study the sensitivity of SimGRL to perturbation strength, we conducted an additional experiment in which we reduced the mixing coefficient from 0.5 to 0.3. This weaker blending introduces milder visual variation during training. Our aim was to test whether SimGRL retains its robustness benefits when the perturbations are less pronounced. Lowering the blending coefficient from $\alpha = 0.5$ to $\alpha = 0.3$ strengthens the presence of the original DMControl frames relative to the overlaid background. Empirically, this adjustment slightly improves the stability of M-SRO across all three seeds and evaluation modes. For example, on the Video Hard benchmark (Table 6), the average reward increases when using $\alpha = 0.3$, suggesting that a weaker background overlay can reduce visual interference while still providing enough perturbation to promote generalization. A similar pattern appears on Train and Video Easy, where $\alpha = 0.3$ produces consistently higher or comparable performance relative to $\alpha = 0.5$. These results indicate that overly strong background blending may introduce unnecessary distractors, while a moderate blend preserves task-relevant cues more effectively.